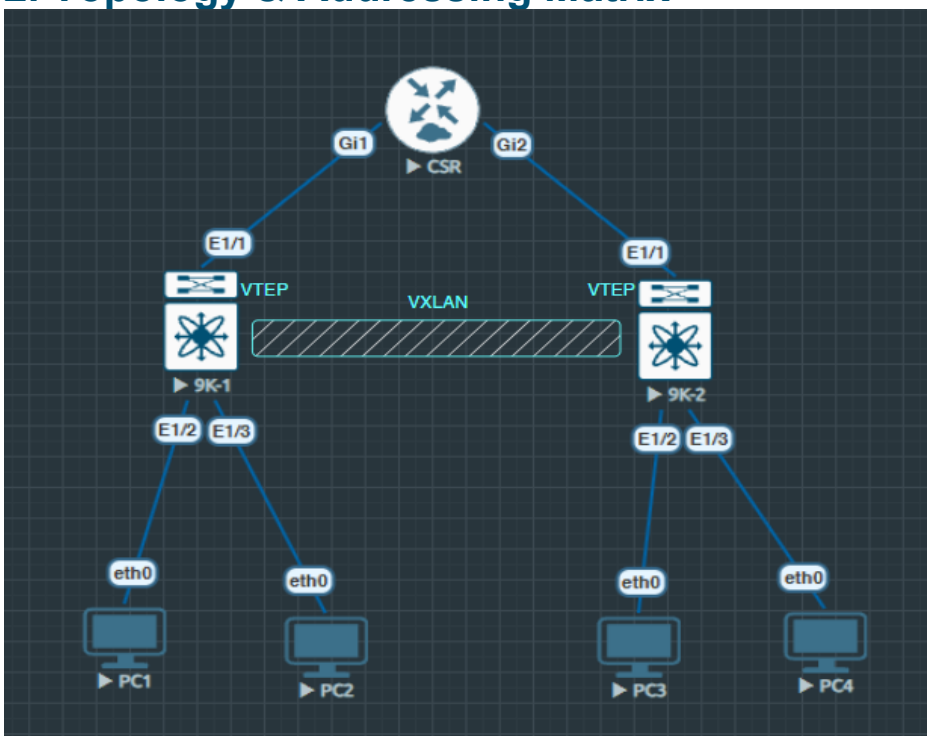
My Project

VXLAN Flood-and-Learn with Multicast Underlay

1. Project Overview & Objectives

- **Objective:** Build a functional VXLAN overlay fabric using a multi-vendor/emulated topology (CSR1000V Spine + Nexus 9000v VTEPs) to stretch Layer 2 broadcast domains across a Layer 3 IP underlay.
- **Core Mechanisms:**
  - **Underlay Routing:** OSPF Area 0 for full IP reachability.
  - **BUM Replication:** PIM Sparse-Mode anchored to a central Rendezvous Point (RP) for multicast-based flood-and-learn replication.
  - **Overlay Binding:** NVE encapsulation mapping local VLANs to global VNIs.

2. Topology & Addressing Matrix



Device	Role / Function	Loopback 0 IP	Uplink Interface	Uplink IP	Connected To
CSR1	Spine / PIM RP	10.1.1.100/32	GigabitEthernet1 GigabitEthernet2	10.0.0.1/30 10.0.0.5/30	9K1 (Eth1/1) 9K-2 (Eth1/1)
9K1	VTEP / Leaf 1	10.1.1.1/32	Ethernet1/1	10.0.0.2/30	CSR1 (Gi1)
9K-2	VTEP / Leaf 2	10.1.1.2/32	Ethernet1/1	10.0.0.6/30	CSR1 (Gi2)

3. Phase 1: Spine & Underlay Setup (CSR1)

Enables IP routing, OSPF, and multicast distribution across the core backbone.

```
! Enable Distributed Multicast Routing
ip multicast-routing distributed

! Rendezvous Point Configuration
ip pim rp-address 10.1.1.100

! Core Loopback
interface Loopback0
 ip address 10.1.1.100 255.255.255.255
 ip pim sparse-mode
 ip ospf 1 area 0

! Uplink to Leaf 1 (9K1)
interface GigabitEthernet1
 ip address 10.0.0.1 255.255.255.252
 ip pim sparse-mode
 ip ospf 1 area 0
 no shutdown

! Uplink to Leaf 2 (9K-2)
interface GigabitEthernet2
 ip address 10.0.0.5 255.255.255.252
 ip pim sparse-mode
 ip ospf 1 area 0
 no shutdown

router ospf 1
```

4. Phase 2: Leaf Switch Configurations (Nexus 9K1 & 9K-2)

A. Global Features & Hardware Carving (Run on both N9K switches)

```
feature ospf
feature pim
feature interface-vlan
feature vn-segment-vlan-based
feature nv overlay

! Free up TCAM space for double-wide ARP/Ethernet entries
hardware access-list tcam region racl 512
hardware access-list tcam region arp-ether 256 double-wide
```

B. Leaf 1 Running Configuration (9K1)

```
hostname 9K1

ip pim rp-address 10.1.1.100 group-list 224.0.0.0/4

vlan 10,20
vlan 10
  vn-segment 100000
vlan 20
  vn-segment 200000

interface Vlan10
  no shutdown
  ip address 10.10.10.1/24

interface Vlan20
  no shutdown
  ip address 10.20.20.1/24

interface nve1
  no shutdown
  source-interface loopback0
 member vni 100000
    mcast-group 239.1.1.1
  member vni 200000
    mcast-group 239.1.1.2

interface Ethernet1/1
  no switchport
  ip address 10.0.0.2/30
  ip router ospf 1 area 0.0.0.0
  ip pim sparse-mode
  no shutdown

interface Ethernet1/2
  switchport access vlan 10

interface Ethernet1/3
  switchport access vlan 20

interface loopback0
  ip address 10.1.1.1/32
  ip router ospf 1 area 0.0.0.0
  ip pim sparse-mode

router ospf 1
```

C. Leaf 2 Running Configuration (9K-2)

Mirrors 9K1 with unique loopback and uplink addressing.

```
hostname 9K-2

ip pim rp-address 10.1.1.100 group-list 224.0.0.0/4
```

```
vlan 10,20
vlan 10
  vn-segment 100000
vlan 20
  vn-segment 200000

interface Vlan10
  no shutdown
  ip address 10.10.10.1/24

interface Vlan20
  no shutdown
  ip address 10.20.20.1/24

interface nve1
  no shutdown
  source-interface loopback0
  member vni 100000
    mcast-group 239.1.1.1
  member vni 200000
    mcast-group 239.1.1.2

interface Ethernet1/1
  no switchport
  ip address 10.0.0.6/30
  ip router ospf 1 area 0.0.0.0
  ip pim sparse-mode
  no shutdown

interface Ethernet1/2
  switchport access vlan 10

interface Ethernet1/3
  switchport access vlan 20

interface loopback0
  ip address 10.1.1.2/32
  ip router ospf 1 area 0.0.0.0
  ip pim sparse-mode

router ospf 1
```

5. Phase 3: Verification & Operational Commands

- **Verify VNI Mapping & Multicast Groups:**

```
show nve vni

Interface VNI      Multicast-group   State Mode Type [BD/VRF]      Flags
--------- -------- ----------------- ----- ---- ------------------ -----
nve1      100000   239.1.1.1         Up    DP   L2 [10]
nve1      200000   239.1.1.2         Up    DP   L2 [20]
```

- **Confirm VLAN-to-VNI State:**

```
show vxlan
Vlan            VN-Segment
====            ==========
10              100000
20              200000
```

- **Verify Multicast Routing State (on CSR1 Spine):**

```
show ip mroute
(*, 239.1.1.1), 00:12:26/00:02:51, RP 10.1.1.100, flags: S
  Incoming interface: Null, RPF nbr 0.0.0.0
  Outgoing interface list:
    GigabitEthernet2, Forward/Sparse, 00:12:14/00:02:51
    GigabitEthernet1, Forward/Sparse, 00:12:26/00:02:48

(10.1.1.2, 239.1.1.1), 00:11:38/00:02:51, flags:
  Incoming interface: GigabitEthernet2, RPF nbr 10.0.0.6
  Outgoing interface list:
    GigabitEthernet1, Forward/Sparse, 00:11:38/00:02:48

(10.1.1.1, 239.1.1.1), 00:11:42/00:02:47, flags:
  Incoming interface: GigabitEthernet1, RPF nbr 10.0.0.2
  Outgoing interface list:
    GigabitEthernet2, Forward/Sparse, 00:11:42/00:02:51

(*, 239.1.1.2), 00:12:26/00:02:52, RP 10.1.1.100, flags: S
  Incoming interface: Null, RPF nbr 0.0.0.0
  Outgoing interface list:
    GigabitEthernet2, Forward/Sparse, 00:12:14/00:02:52
    GigabitEthernet1, Forward/Sparse, 00:12:26/00:02:49

(10.1.1.2, 239.1.1.2), 00:11:37/00:02:52, flags:
  Incoming interface: GigabitEthernet2, RPF nbr 10.0.0.6
  Outgoing interface list:
    GigabitEthernet1, Forward/Sparse, 00:11:37/00:02:49

(10.1.1.1, 239.1.1.2), 00:11:41/00:02:49, flags:
  Incoming interface: GigabitEthernet1, RPF nbr 10.0.0.2
  Outgoing interface list:
    GigabitEthernet2, Forward/Sparse, 00:11:41/00:02:52

(*, 224.0.1.40), 00:13:57/00:02:10, RP 10.1.1.100, flags: SJCL
  Incoming interface: Null, RPF nbr 0.0.0.0
  Outgoing interface list:
    Loopback0, Forward/Sparse, 00:13:56/00:02:10
```